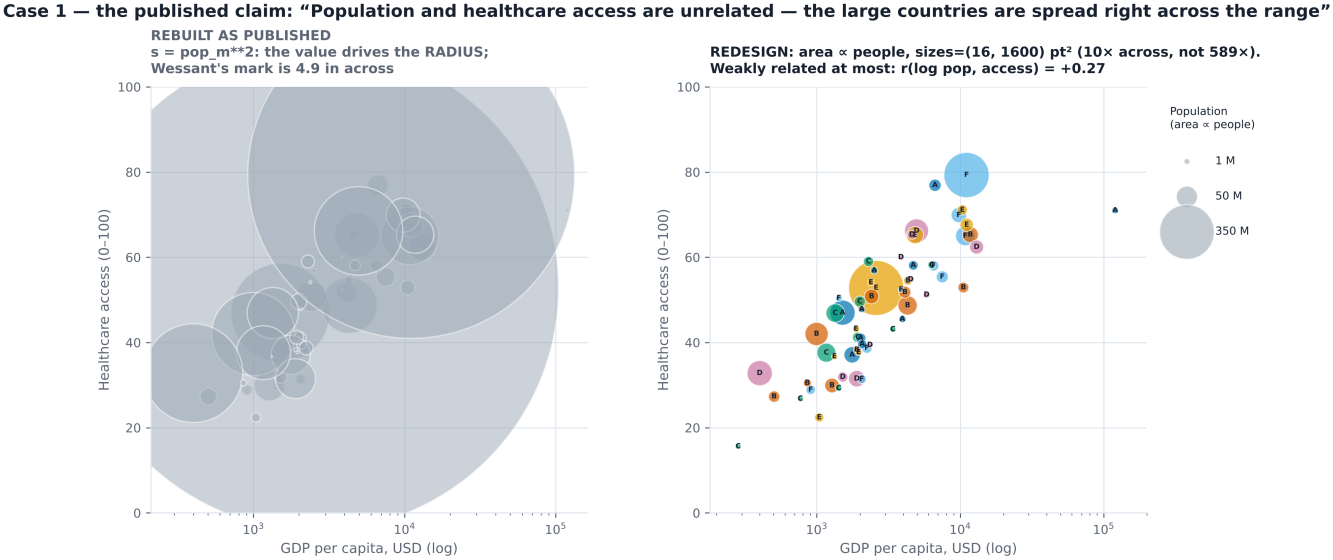
Week 5 · Part A — nine charts, taken apart

Qurat ul Aien · Roll No. AI-016 · CDB601220 Data Visualization · Week 5

Each case: the faithful rebuild (left) and the redesign (right), then the conclusion and decoy, the lying channel or boundary with the number that proves it, and the redesign. Self-checks are asserted in *partA_charts.py*; every number is printed in *partA_output.txt*. All data is synthetic course data generated for CDB601220.

Case 1 · “Population and healthcare access are unrelated — the large countries are spread right across the range”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (health_indicators.csv, 60 countries).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

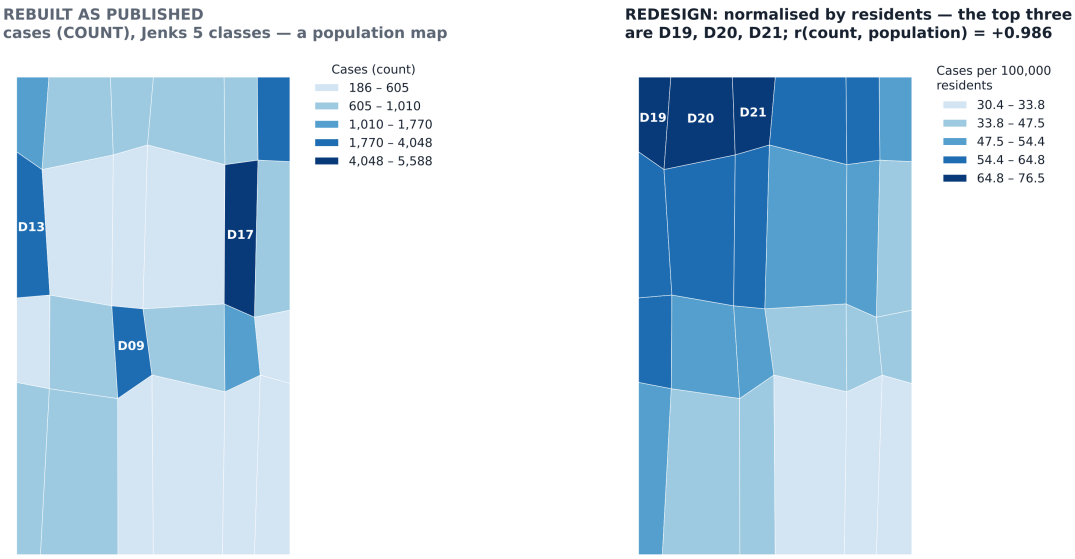
2 · Conclusion and decoy. Conclusion invited: a country's size tells you nothing about its access. **Decoy:** 'the log x-axis distorts the picture'. Ruled out: GDP spans $283–$120,219 (425×), a multiplicative spread, so log is the correct axis — the x and y positions are fine.

3 · Mechanism. A value driving the mark's radius instead of its area. *s* is area in points², so *s = pop_m**2* makes the **diameter** proportional to population: 589× across (area 346,921×). Wessant's mark is 353 pt = **4.9 in** wide on a 6.4 in figure and buries the others, while the 9 countries under 2.0 M are under 2 pt across (3 under 1 pt). *Finding: the brief's '589× the area' reproduces as 589× the diameter.* With the size channel repaired the relation is weak but positive: r(log pop, access) = +0.27; countries ≥ 50 M average 53.4 vs 47.7.

4 · Redesign. Area ∝ population with sizes=(16, 1600) pt² (10× across), size legend with real values (1, 50, 350 M), region hue plus letter.

Case 2 · “The outbreak is concentrated in the three districts around the capital”

Case 2 — the published claim: “The outbreak is concentrated in the three districts around the capital”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (district_health.csv + districts.geojson; Lambert cylindrical equal-area).

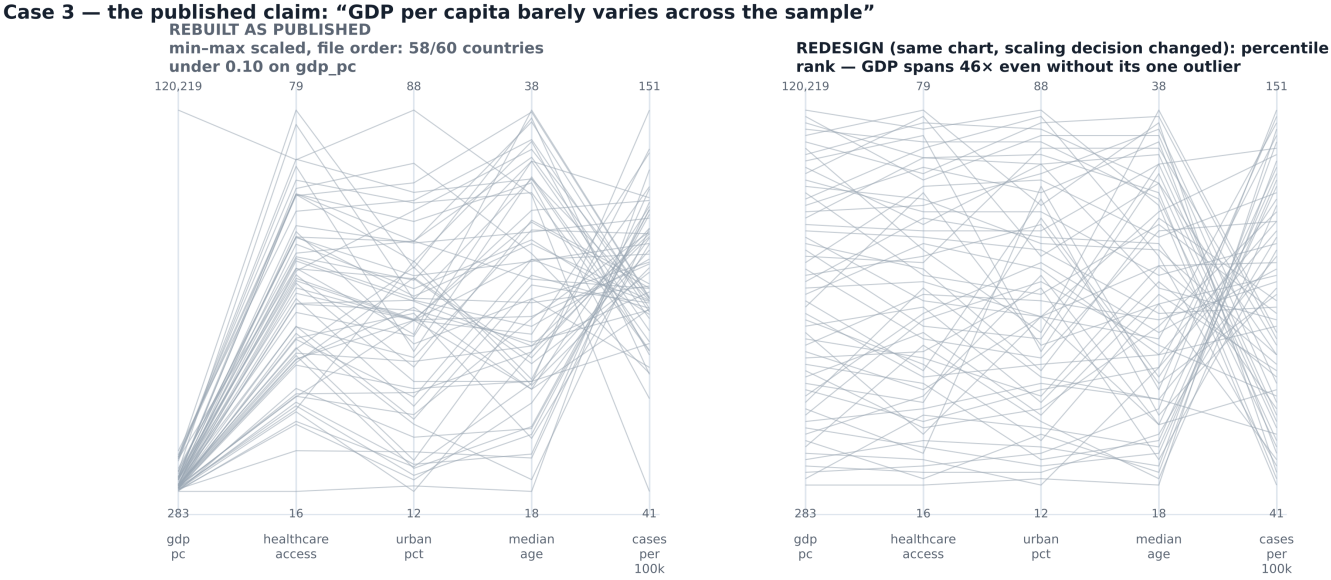
1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

2 · Conclusion and decoy. Conclusion invited: D17, D13 and D09 are where the risk is. **Decoy:** 'the class scheme put them there'. Ruled out: equal-interval and Jenks put D17 alone in the top count class and quantile puts D09/D11/D13/D17/D24 there — every scheme ranks the same big districts, because the ranking is by count.

3 · Mechanism. A raw-count choropleth is a population map. r(cases, population) = **+0.986**, r(rate, population) = **−0.000**. The count top three hold 6.3–11.0 M people each (41.8% of residents, 42.8% of cases) and their rates are 49.9, 64.0 and 51.0 against a national 52.8. Top three by rate: **D19, D20, D21** — overlap with the count list: **zero**.

4 · Redesign. The same map, normalised: cases per 100,000 residents, legend titled with the denominator.

Case 3 · “GDP per capita barely varies across the sample”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (health_indicators.csv, 60 countries; axis ends = real min/max).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

2 · Conclusion and decoy. Conclusion invited: income is roughly equal everywhere. **Decoy:** 'GDP genuinely varies little'. Ruled out: GDP has the highest coefficient of variation of the five columns (**2.65** vs 0.19–0.40), and even without its largest value it spans $283–$13,009 (**46×**).

3 · Mechanism. Min–max scaling flattened by one outlier. Aldonia's $120,219 is **9.2×** the next country, so the min–max denominator is set by one row: **58 of 60** countries land under 0.10 and the median at **0.017**; the other 59 share 10.6% of the axis. z-scoring has the same problem (Aldonia +7.5 sd).

4 · Redesign. Same parallel coordinates chart, one decision changed: percentile-rank scaling (median 0.508; the 59 use 98% of the axis). *Not fixed by changing the chart type.*

Case 4 · “Healthcare access and case rate are unrelated — there is no crossing pattern between them”

Case 4 — the published claim: “Healthcare access and case rate are unrelated — there is no crossing pattern between them”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (health_indicators.csv; both panels min–max so only the ORDER differs).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

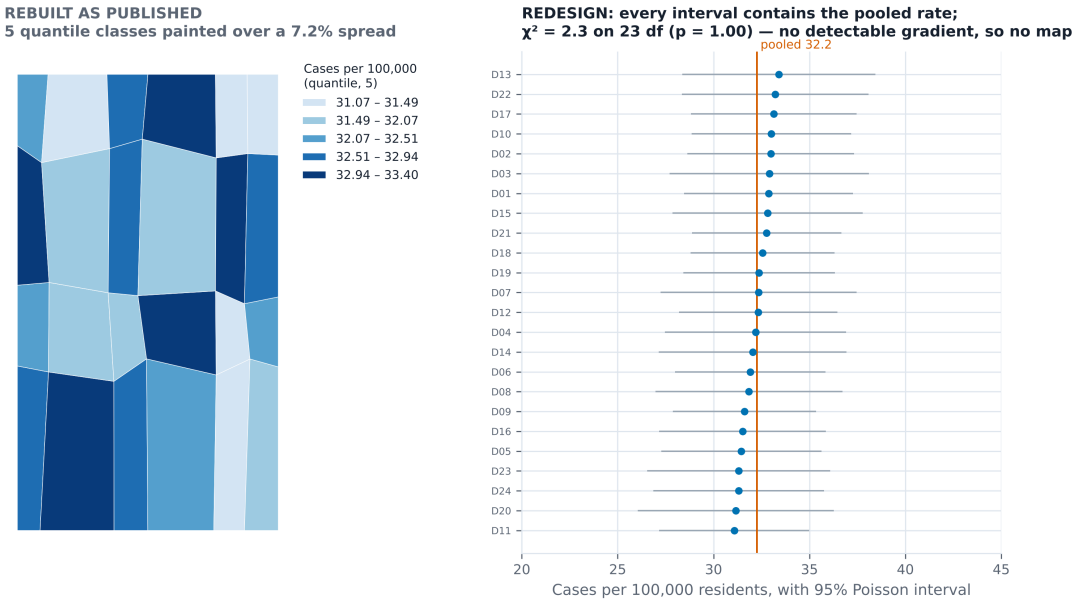
2 · Conclusion and decoy. Conclusion invited: no link between access and cases. **Decoy:** 'n = 60 is too small to show a pattern'. Ruled out: r = −0.80 with n = 60 has p = 1.9×10^{-14}.

3 · Mechanism. An axis order that separates the related pair. Crossings are only readable between ADJACENT axes; in file order access and cases are **three apart**. Placed side by side, **81%** of line pairs cross (an X, r = **−0.799**); the four file-order gaps show only 15%, 21%, 28% and 71%.

4 · Redesign. Same chart, same min–max scaling, order decision changed: greedy |r| chain access → cases → urban → median age → GDP. *Not fixed by changing the chart type.*

Case 5 · “There is a clear five-level gradient in incidence across the federation”

Case 5 — the published claim: “There is a clear five-level gradient in incidence across the federation”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (case05_flat_rates.csv + districts.geojson).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

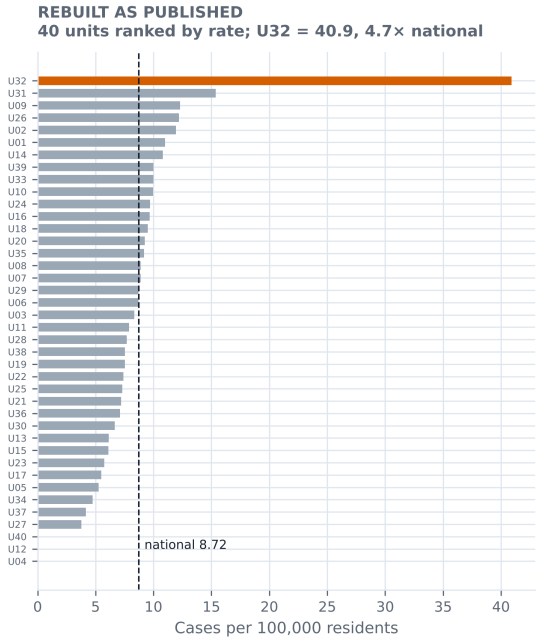
2 · Conclusion and decoy. Conclusion invited: risk rises steadily across the map. **Decoy:** 'just use Jenks or equal interval instead'. Ruled out: all three schemes still paint five confident colours (breaks within 31.07–33.40) — no scheme can say whether a pattern exists.

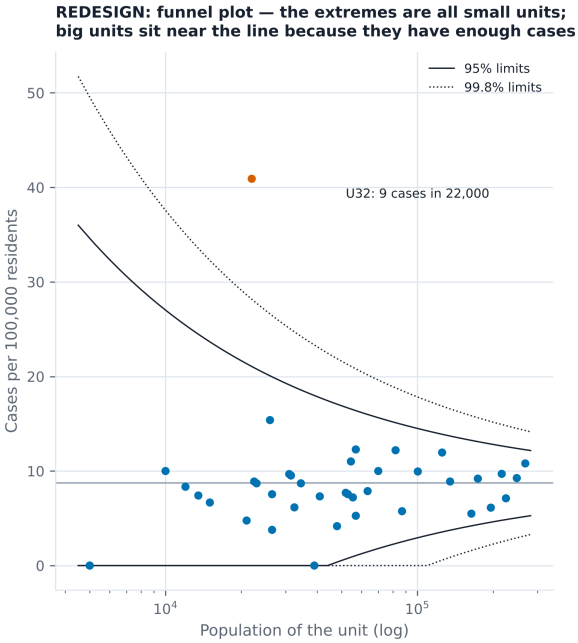
3 · Mechanism. Five quantile classes painted over a 7.2% spread. Rates run 31.07–33.40 per 100,000, a spread of **7.2%** of the mean (2.33 per 100,000), while each district's 95% Poisson interval is ±3.7 to ±5.2 and **all 24** contain the pooled 32.2. A χ^2 homogeneity test gives **2.3 on 23 df (p = 1.00)** — less variation than chance alone would produce.

4 · Redesign. A dot plot of the 24 rates with 95% intervals and the pooled line: no detectable gradient, so no map.

Case 6 · “Unit U32 is a hotspot — it is running at nearly five times the national rate”

Case 6 — the published claim: “Unit U32 is a hotspot — it is running at nearly five times the national rate”





Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (smallarea_cases.csv, 40 units sharing one true rate).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

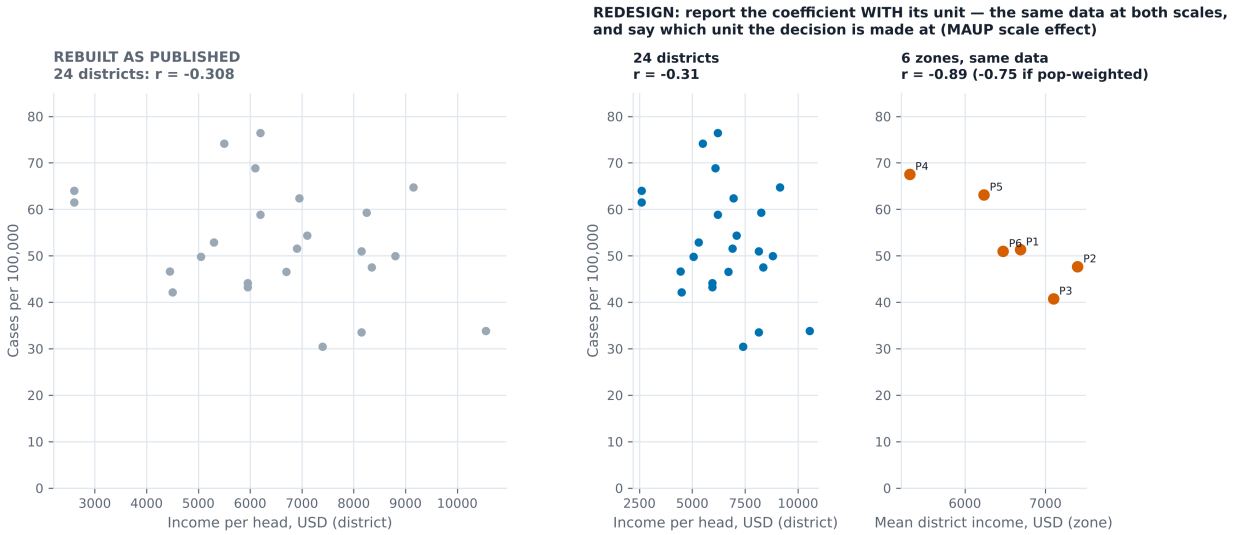
2 · Conclusion and decoy. Conclusion invited: U32 needs intervention. **Decoy:** 'U32 really has an outbreak'. The file states all 40 units share one true rate; and every other 'high' unit (15.4, 12.3, 12.2…) is a small unit inside the funnel.

3 · Mechanism. A small-denominator rate read as a hotspot. U32's 40.9 (**4.7×** the pooled 8.72) comes from **9 cases in 22,000 people**; its exact 95% interval is 18.7–77.7 (a 4.2-fold range). The 10 units over 100,000 people all sit at 5.5–12.0; the 10 units under 25,000 span 0–40.9. U32 is outside the 99.8% funnel (P = 0.0002), which with 40 units happens on about 1 map in 143 — a rare extreme, not a measured rate.

4 · Redesign. A funnel plot (rate vs population, 95% and 99.8% Poisson limits). Rule: flag U32 for follow-up, merge units under ~43,000 people before mapping.

Case 7 · “District income and case rate are only weakly related — r is about −0.31”

Case 7 — the published claim: “District income and case rate are only weakly related — r is about −0.31”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (district_health.csv; zones = districts_coarse.geojson `members`).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

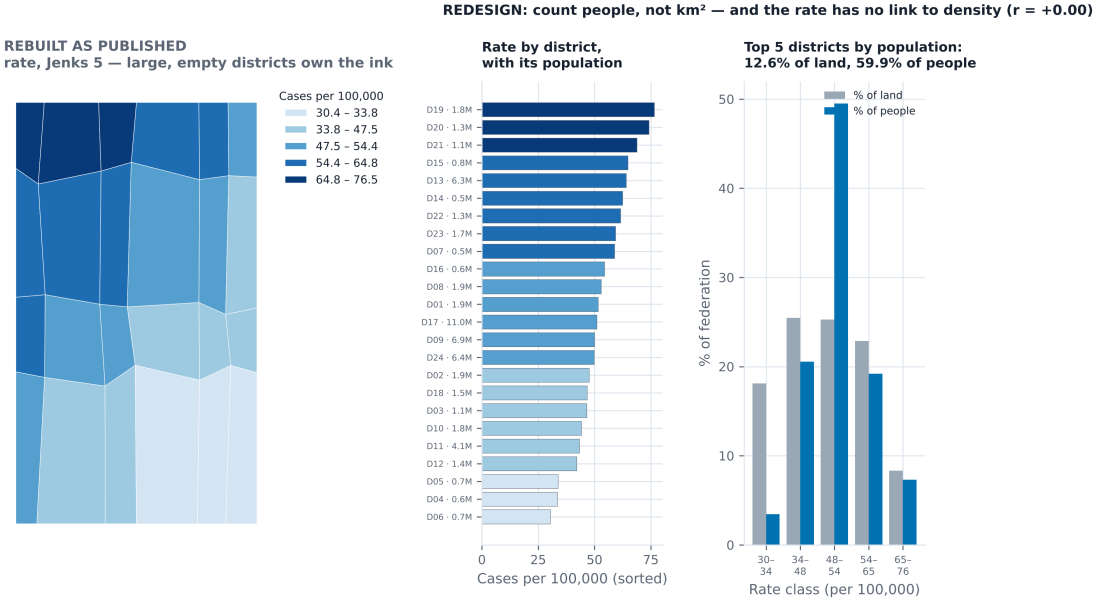
2 · Conclusion and decoy. Conclusion invited: income barely matters. **Decoy:** 'n = 24 is too small for a reliable r'. Ruled out as the cause: the SAME people give very different r at different scales, which n cannot explain.

3 · Mechanism. A correlation quoted at the wrong areal scale (MAUP). Across 24 districts r = **−0.308** (p = 0.14); aggregated to the 6 zones of districts_coarse, r = **−0.889** (−0.746 with population-weighted income). Within zones it ranges from −0.69 to +0.77. The coefficient is a property of the units, not only of income and disease; and a zone-level r says nothing about individuals (ecological fallacy).

4 · Redesign. Both scales side by side, each r labelled with its unit; state the scale the decision is made at.

Case 8 · “This is a rural outbreak — look how much of the map is dark”

Case 8 — the published claim: “This is a rural outbreak — look how much of the map is dark”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (district_health.csv + districts.geojson, equal-area).

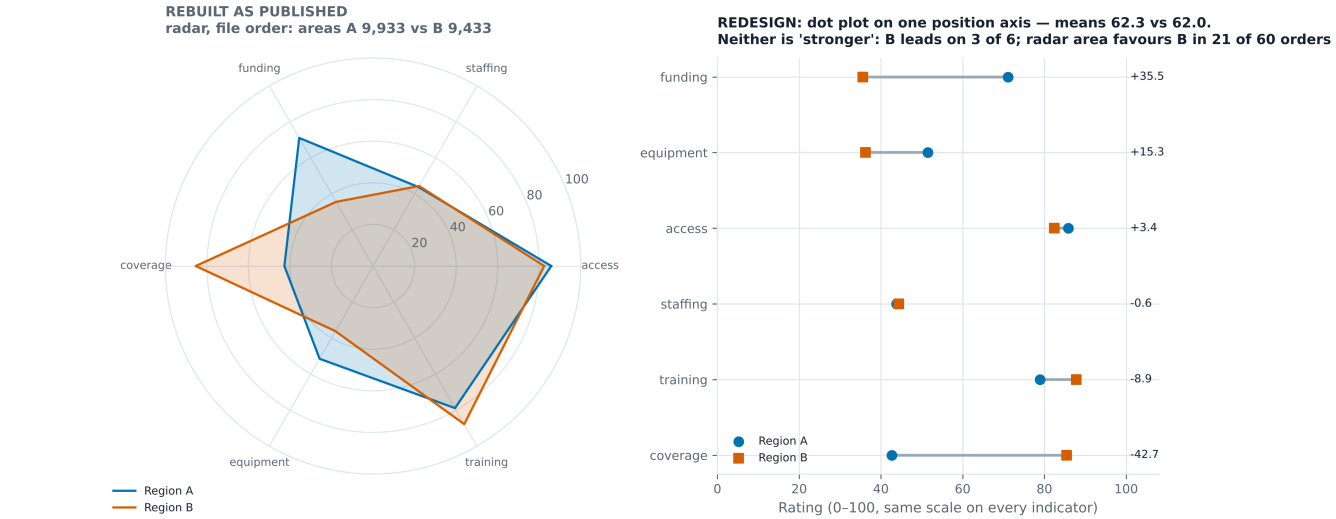
1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

2 · Conclusion and decoy. Conclusion invited: risk lives in the big, empty districts. **Decoy:** 'the projection inflates the north'. Ruled out: the map is Lambert equal-area (error < 10^{-10} %); even Mercator would inflate north vs south by only 1.21× (Part B4).

3 · Mechanism. Area read as people. The five most populous districts are **12.6% of the land and 59.9% of the people**; the top two rate classes cover 31.2% of land but 26.5% of people. The middle class holds **49.5% of residents on 25.3% of land**. And the rate has no link to density at all: r(rate, density) = **+0.001** (+0.14 on log density) — nothing here is 'rural'.

4 · Redesign. A sorted bar chart of rate with each district's population, beside land-share vs people-share for every rate class.

Case 9 · “Region A is the stronger of the two — its profile encloses visibly more area”



Source: CDB601220 Week 5 synthetic course data — fictional places, not real ones (region_profiles.csv, six 0–100 ratings).

1 · Rebuild. Recipe followed exactly; self-check reproduced and asserted.

2 · Conclusion and decoy. Conclusion invited: A outperforms B. **Decoy:** 'the six axes are on different scales'. Ruled out: all six indicators are 0–100.

3 · Mechanism. A radar chart whose enclosed area depends on axis order. The means are **62.3 vs 62.0** (difference 0.33), but area multiplies NEIGHBOURING values: file order gives 9,933 vs 9,433, and across the 60 distinct orders A − B ranges from −1,131 to +1,659 — B encloses more in **21 of 60**. B leads on 3 of 6 indicators (coverage by 42.7, funding favours A by 35.5).

4 · Redesign. A dot plot (dumbbell) on one position axis, sorted by the difference, with the six gaps printed.